NAGA VENKATA ANVITHA ACHYUTA

AI/ML Engineer
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SUMMARY

Machine Learning Engineer with 4+ years of experience building scalable GenAI, LLM inference, and real-time ML systems for enterprise applications. Skilled in distributed AI infrastructure, RAG pipelines, streaming architectures, and MLOps using Kubernetes, Kafka, Flink, Ray Serve, and Triton Inference Server. Experienced in developing production-grade AI platforms, anomaly detection systems, and low-latency inference services processing high-volume enterprise workloads. Strong background in AI observability, model serving, and deploying reliable machine learning systems across AWS and GCP cloud environments.

PROFESSIONAL EXPERIENCE

Scale AI — **Dec 2024 – Present**
Machine Learning Engineer — San Francisco, CA

- Improved real-time AI decisioning latency by 45% by redesigning distributed LLM inference pipelines using Kubernetes autoscaling and request batching across multi-GPU clusters supporting enterprise GenAI workloads.
- Delivered a production-grade AI orchestration system enabling multi-step reasoning across streaming data sources, reducing manual analyst intervention by 60% in enterprise decision workflows.
- Built scalable AI evaluation and observability framework for GenAI systems, increasing model reliability detection coverage by 38% and significantly reducing production hallucination incidents in enterprise deployments.
- Designed and deployed event-driven AI microservices architecture using Kafka, Flink, and gRPC to power real-time LLM inference pipelines processing millions of daily requests with sub-second latency.
- Built RAG (Retrieval-Augmented Generation) infrastructure using vector databases (Pinecone/Weaviate/FAISS) and hybrid search systems (Elasticsearch/OpenSearch) to support enterprise knowledge copilots.
- Developed LLM serving layer using FastAPI, Ray Serve, and Triton Inference Server with intelligent workload scheduling for scalable enterprise AI inference.
- Engineered AI workflows using LangChain/LlamaIndex-style orchestration for planning, retrieval, and execution pipelines supporting autonomous enterprise task automation.
- Implemented real-time feature engineering pipelines with Redis, Kafka Streams, and Spark Streaming for personalization and contextual AI responses in enterprise GenAI applications.
- Built cloud-native MLOps infrastructure on Kubernetes and Terraform enabling automated model deployment, versioning, rollback, and GPU autoscaling across multi-cloud environments (AWS/GCP).
- Developed frontend + API-driven AI dashboards (React + TypeScript + WebSockets) for monitoring AI system health, prompt flows, latency metrics, and model performance in real tim

Accenture — **Dec 2020 - July 2023**
Machine Learning Engineer — India

- Improved incident detection accuracy by 45% by building real-time anomaly detection pipelines using streaming telemetry data from distributed cloud infrastructure for production AIOps systems.
- Reduced Mean Time To Resolution (MTTR) by 60% by designing AI-driven root-cause analysis systems that correlated logs, metrics, and traces across microservices using graph-based dependency modeling and predictive ML signals.
- Reduced cloud operational costs by 40% by implementing auto-scaling intelligence and predictive workload optimization models integrated into Kubernetes-based infrastructure orchestration pipelines.
- Designed and deployed real-time ML inference microservices using FastAPI, Spring Boot, and gRPC, enabling low-latency (<100ms) prediction APIs for AIOps anomaly detection and alert classification systems.
- Built high-throughput event streaming pipelines using Apache Kafka, Apache Flink, and Spark Structured Streaming to process millions of telemetry events per minute from distributed enterprise cloud systems.
- Developed and productionized ML models (XGBoost, LSTM, Transformer-based architectures) for time-series forecasting, log anomaly detection, and predictive incident management in large-scale IT environments.
- Implemented full MLOps lifecycle automation using Docker, Kubernetes, MLflow, and CI/CD pipelines (Jenkins/GitHub Actions) for model versioning, canary deployments, and automated retraining workflows.
- Integrated observability and monitoring systems using Prometheus, Grafana, ELK Stack, and Splunk, enabling real-time model drift detection, system health tracking, and performance optimization across distributed AI services.

TECHNICAL SKILLS

Programming & Frameworks: Python, Java, SQL, FastAPI, Spring Boot

Machine Learning & AI: Machine Learning, Deep Learning, Generative AI, LLM Inference Optimization, RAG (Retrieval-Augmented Generation), Time-Series Forecasting, Anomaly Detection, Predictive Analytics, XGBoost, LSTM, Transformer Models

LLM & AI Systems: vLLM, Ray Serve, Triton Inference Server, LangChain, LlamaIndex, Prompt Engineering, AI Evaluation & Observability, Model Serving, Multi-Agent Orchestration

Data & Streaming Technologies: Apache Kafka, Apache Flink, Spark Streaming, Spark Structured Streaming, Redis, Kafka Streams

Vector Databases & Search: Pinecone, Weaviate, FAISS, Elasticsearch, OpenSearch

Cloud & MLOps: Docker, Kubernetes, Terraform, MLflow, CI/CD, Jenkins, GitHub Actions, GPU Autoscaling, AWS, GCP

APIs & Distributed Systems: REST APIs, gRPC, Microservices Architecture, Distributed ML Systems, Event-Driven Architecture

Monitoring & Observability: Prometheus, Grafana, ELK Stack, Splunk, Model Drift Monitoring, AI System Observability

EDUCATION

Western New England University

Master of Science in Computer Science